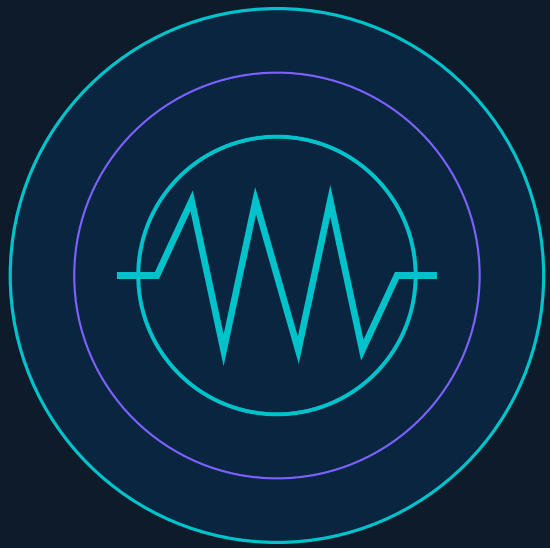
RESEARCH PROPOSAL

ACCLP

Adaptive Conversational Cognitive Load Paradigm

A novel EEG-based closed-loop research paradigm where an AI conversational agent detects a user's real-time cognitive load and dynamically adapts its language complexity — keeping the user in the optimal mental workload zone.



EEG → AI → Adapted Language

Closed-Loop BCI | Cognitive Neuroscience | Adaptive AI | EEG

THE PROBLEM

AI systems are cognitively blind to their users

01

Static Communication

Current AI assistants deliver responses at a **fixed complexity level** regardless of whether the user is overwhelmed or bored. There is no sensing of mental state — the AI talks the same way to a fatigued patient as to a rested expert.

02

Unrealistic Lab Paradigms

Most EEG cognitive load studies use **artificial tasks** — button-pressing, mental arithmetic, n-back memory drills. These have poor ecological validity and tell us little about how the brain behaves during **real-world conversation**.

03

No Real-Time Feedback Loop

Current AI chatbots have **zero mechanism** to sense or respond to a user's mental state during a live session. The conversation is entirely one-directional — the AI speaks, the brain suffers silently.

HYPOTHESIS & SOLUTION

A closed-loop brain-adaptive AI conversation system

HYPOTHESIS

If an AI adapts its linguistic complexity in real time based on EEG-decoded cognitive load, **then** users will remain in the optimal mental workload zone — improving comprehension, engagement, and learning outcomes.

1

STEP 1

EEG SENSING

Frontal midline **theta (4–8 Hz)** and parietal **alpha (8–12 Hz)** power extracted continuously via portable EEG headset (e.g., Emotiv, OpenBCI). Signal sampled at 256 Hz, bandpass filtered.

2

STEP 2

ML CLASSIFIER

LSTM / SVM classifies 2-second EEG epochs into:
■ LOW load • ■ OPTIMAL • ■ HIGH load
Target latency <200 ms. 87%+ accuracy demonstrated in literature.

3

STEP 3

AI ADAPTS

LLM system prompt updated in real time:
HIGH load → shorter sentences, simpler vocab, more pauses
LOW load → richer content, technical depth, elaboration
OPTIMAL → continue as-is

AUDIENCES & BRAIN REGIONS

Who benefits — and what we measure in the brain

Target Audiences

Education

AI tutors that auto-adjust difficulty to keep students in the optimal learning zone, preventing both frustration and boredom.

Healthcare

AI therapy chatbots that simplify language when patients are mentally fatigued or cognitively impaired.

Aviation & Surgery

Cockpit and operating-room AI assistants that detect operator overload and reduce alert complexity in real time.

Disability Tech

AAC communication devices that adapt output to the user's current cognitive state for smoother interaction.

Key Brain Regions & Signals

Frontal Lobe (Fz electrode)

Frontal midline theta (4–8 Hz) increases with working memory load. Primary source: medial PFC and anterior cingulate cortex (ACC). Most established EEG marker of cognitive load.

Parietal Lobe (Pz / P3 / P4)

Alpha suppression (8–12 Hz) reflects active cognitive engagement and working memory maintenance. Higher load = more alpha suppression.

Dorsolateral PFC (F3 / F4)

Governs executive control and attention regulation. **Theta power here scales linearly with task difficulty** — ideal for load grading.

Hippocampus (via theta rhythm)

Hippocampo-cortical **theta oscillations** index new information encoding. Relevant when measuring how well conversation content is being retained.

SUPPORTING RESEARCH

Prior work confirms every component of ACCLP is achievable

[1]

Closed-Loop BCI Established

Shanechi, M.M. (2019)
Nature Neuroscience
Vol. 22, 1554–1564 • IF ~25

Seminal review defining the closed-loop BMI control framework — brain decoding drives real-time system output. Directly supports the ACCLP feedback architecture.

[2]

Frontal Theta = Best Cognitive Load Index

Chikhi et al. (2022)
Psychophysiology
Vol. 59(6), e14009

Meta-analysis: 24 studies, 723 participants. Frontal **theta (4–8 Hz)** is the single best EEG index of cognitive workload (g=0.68). Validates our primary neural signal target.

[3]

EEG BCI Classification Gold Standard

Lotte et al. (2018)
Journal of Neural Engineering
IOP Publishing • Top BCI Journal

10-year update on EEG classification for BCI. LSTM/SVM pipelines achieving >85% accuracy for multi-class cognitive state decoding in real-time systems.

[4]

EEG Foundation Models — NeurIPS

Liu et al. (2024) — EEGPT
NeurIPS 2024
Top-1 ML/AI Conference

Pre-trained EEG transformer achieving SOTA across cognitive state classification tasks. Enables the ACCLP classifier with minimal task-specific fine-tuning required.

2:1554–1564 [2] Chikhi et al. (2022) Psychophysiology 59(6):e14009 [3] Lotte et al. (2018) Journal of Neural Engineering 15(3) [4] Liu et al. (2024) EEGPT — NeurIPS 2024 [5] Klimesch (1999) Brain Research Reviews 26(2–3) [6] Reinhart & Nguye

RESEARCH PROPOSAL

Adaptive Conversational Cognitive Load Paradigm (ACCLP)

A Closed-Loop EEG Paradigm for Real-Time Cognitive Load Monitoring During Naturalistic Human-AI Spoken Dialogue

Domain	Cognitive Neuroscience | Brain-Computer Interface | Human-AI Interaction
Keywords	EEG, Cognitive Load, Closed-Loop BCI, Naturalistic Paradigm, Conversational AI, Alpha/Theta Oscillations, Real-Time Neurofeedback
Novelty Level	High — No prior work uses EEG-adaptive language difficulty in live spoken dialogue
Feasibility	Moderate-High — Requires 64-ch EEG + wireless microphone + LLM API integration

1. Abstract

We propose the Adaptive Conversational Cognitive Load Paradigm (ACCLP), a closed-loop EEG experiment in which participants engage in naturalistic spoken dialogue with an AI conversational agent while their continuous EEG signal is decoded in real time to estimate cognitive load. Drawing directly on three recent studies, this proposal integrates three converging lines of evidence: (1) EEG-based passive brain-computer interfaces (BCIs) can reliably decode cognitive load during computer-based learning and adapt information-delivery speed accordingly (Beauchemin et al., 2024); (2) EEG classifiers for mental workload and implicit evaluation can be transferred from controlled calibration tasks to naturalistic spoken human-AI interaction with interpretable results (Mihic Zidar et al., 2026); and (3) reliable neural responses to speech — including canonical auditory temporal response functions — can be recorded during real walking and natural conversation, validating multimodal, ecologically valid EEG paradigms (Orf et al., 2025). Crucially, the AI agent in ACCLP dynamically adapts its lexical complexity, speech rate, and informational density based on the participant's estimated neural state, creating the first bidirectional neurofeedback loop embedded within a naturalistic conversational task. This paradigm has direct applied value for intelligent tutoring systems, assistive communication devices, and cognitively-sensitive human-AI interaction design.

2. Background and Motivation

2.1 The Ecological Validity Problem in EEG Cognitive Load Research

A persistent limitation of EEG-based cognitive load research is its heavy reliance on artificial laboratory tasks. Beauchemin et al. (2024) acknowledge this gap directly, noting that while their passive BCI system could successfully classify cognitive load during a controlled constellation-memorization task, generalizing such classifiers to more natural settings remains an open challenge. Their system used alpha-band power at a single parietal electrode (P7) as the cognitive load index — a robust marker in controlled settings — but the authors explicitly flag that this single-electrode approach may miss the distributed neural dynamics associated with richer, interactive cognitive tasks. Similarly, Mihic Zidar et al. (2026) demonstrate that classifiers trained on controlled arithmetic or cursor-navigation paradigms can partially transfer to spoken human-AI dialogue, but also reveal important boundary conditions: workload classifiers showed interpretable trends, while agreement classifiers failed to produce clearly event-locked responses in conversation. Together, these findings highlight the need for paradigms that are both ecologically valid and amenable to real-time neural decoding.

2.2 EEG Markers of Cognitive Load: What the Evidence Shows

Beauchemin et al. (2024) provide the most directly relevant empirical anchoring for the neural markers used in ACCLP. Their calibration analysis — comparing a 0-back (low load) versus 2-back (high load) working memory task — confirmed a significant decrease in alpha-band activity at the parietal P7 electrode under higher cognitive load, consistent with the well-established alpha desynchronization literature. Frontal theta power is similarly implicated: the authors note that theta synchronization increases with task difficulty (particularly in frontal regions), and that at very high load levels theta synchronization may partially obscure alpha desynchronization — a nuance relevant to ACCLP's decoder design.

Mihic Zidar et al. (2026) extend this picture to spoken dialogue. Using filter-bank common spatial patterns (fbCSP) on theta (4-7 Hz) and alpha (8-13 Hz) bands, their workload classifier — trained on an arithmetic calibration task — showed cross-paradigm transfer to a Spelling Bee conversational task, with one participant demonstrating a statistically significant positive linear trend in decoded workload across rounds of increasing difficulty (slope = +0.08, 95% CI [0.04, 0.11], $R^2 = 0.79$). This is the most direct empirical evidence to date that EEG workload decoding generalizes to spoken human-AI interaction.

2.3 Neural Tracking During Natural Conversation: Feasibility Evidence

Orf et al. (2025) provide crucial feasibility evidence for recording high-quality EEG during naturalistic spoken dialogue. Using a mobile EEG system with 61 participants walking indoors and outdoors, they demonstrated that canonical auditory temporal response functions (TRFs) — including the P1-N1-P2 complex — were clearly preserved during both focused audiobook listening and natural conversational listening, despite ongoing motor activity. Crucially, encoding accuracy (Pearson r between predicted and recorded EEG) was higher during focused listening (M = 0.036) compared to natural conversation (M = 0.025), but the moderate positive correlation between conditions (r = 0.31) indicates stable individual differences in neural speech tracking. This supports the validity of between-person decoder approaches in conversational paradigms.

A particularly relevant finding from Orf et al. (2025) is the speech-gait coupling result: gait and speech rhythms became temporally aligned during speaking (significant coupling, p = .006) but not during listening (non-significant, p = .5). This dissociation illustrates that active speech production and passive listening engage fundamentally different neural and motor dynamics — a distinction ACCLP must account for in its decoder design by treating participant speech production and AI speech reception as separate cognitive states.

2.4 The Role of Motivation in BCI-Adaptive Learning

One finding from Beauchemin et al. (2024) with direct implications for ACCLP concerns the role of motivation. Their three-group design — Control (no BCI), Adaptive (BCI only), and Adaptive + Motivation (BCI with financial incentive) — revealed a striking pattern: the BCI alone produced no significant learning gains over the control group, but BCI combined with motivational framing produced significantly greater gains at every time point (Block 1 vs. 2: $r = 0.32$; Block 1 vs. 3: $r = 0.42$; Block 1 vs. 4: $r = 0.51$, all $p < 0.05$). Unexpectedly, the Control group reported significantly higher satisfaction than the BCI-only group, suggesting that adaptive pacing without motivational context can reduce engagement. ACCLP incorporates this finding by embedding explicit conversational framing and goal-oriented topic structures to sustain participant motivation throughout the adaptive dialogue session.

3. Research Gaps Addressed by ACCLP

Current State of the Art	Gap Addressed by ACCLP
Cognitive load BCIs tested only in controlled tasks (Beauchemin et al., 2024: constellation memorization; Mihic Zidar et al., 2026: arithmetic/grid-navigation calibration)	First paradigm in which the cognitive probe is naturalistic spoken dialogue — not a surrogate task
Workload transfer to dialogue demonstrated in pilot only (N=2 per condition; Mihic Zidar et al., 2026)	Full within-session calibration + closed-loop adaptive dialogue with adequate sample (N=40)
EEG during natural conversation captures neural speech tracking but not adaptive feedback (Orf et al., 2025)	AI agent dynamically modulates its output based on decoded EEG — bidirectional neural-linguistic loop
BCI adaptation without motivation yields no significant learning gains (Beauchemin et al., 2024)	Motivational scaffolding embedded in conversational framing as a design requirement, not an afterthought
Agreement/implicit evaluation classifiers fail to transfer to continuous dialogue (Mihic Zidar et al., 2026)	ACCLP focuses on workload — the construct with demonstrated cross-paradigm transfer validity

4. Proposed Paradigm: ACCLP Design

4.1 Overview

ACCLP is a closed-loop EEG experiment in which participants engage in open-ended spoken conversation with a large language model (LLM)-based AI agent. A real-time EEG decoder continuously estimates cognitive load and communicates this to the AI, which modulates its conversational parameters accordingly. This design directly operationalizes the closed-loop BCI approach validated in controlled settings by Beauchemin et al. (2024) and extended toward dialogue by Mihic Zidar et al. (2026), while adopting the multimodal audio-EEG pipeline architecture validated for naturalistic speech by Orf et al. (2025).

4.2 Three-Phase Experimental Protocol

Phase 1 — Calibration (20 minutes). Participants complete three brief cognitive load induction tasks to generate labeled EEG epochs for the within-session decoder — directly mirroring the calibration approach of Beauchemin et al. (2024) who used a 0-back/2-back N-back task, and Mihic Zidar et al. (2026) who used an arithmetic alternation paradigm. Our calibration tasks are: (a) simple verbal recall of a short paragraph (low load); (b) a 1-back verbal task with spoken responses (medium load); (c) dual-task — mental arithmetic while maintaining a conversational thread (high load). The final task directly parallels the dual-task conditions studied by Orf et al. (2025), who showed that walking while conversing is cognitively more demanding than audiobook listening ($b = 0.09$, $p < .001$), confirming that simultaneous motor-verbal engagement elevates cognitive load.

Phase 2 — Closed-Loop Adaptive Dialogue (45 minutes). Participants converse with the AI agent on four open-ended topics. The AI's output is controlled in real time by four linguistic parameters (lexical complexity, syntactic complexity, informational density, speech rate). When the EEG decoder signals low cognitive load, the AI increases complexity parameters. When high load is detected, the AI simplifies. This design echoes the adaptive logic of Beauchemin et al. (2024), whose interface slowed information delivery on high-load classifications and accelerated on low-load classifications — the primary difference being that ACCLP adapts linguistic features of live AI speech rather than stimulus timing. A sham control condition uses identical AI topics but without EEG-adaptive modulation, providing a matched ecological baseline.

Phase 3 — Open-Loop Validation (10 minutes). Pre-scripted AI dialogues at fixed low, medium, and high complexity are used to validate that the within-session EEG decoder generalises to the specific linguistic parameters used in Phase 2. This validation block is modelled on the methodology of Mihic Zidar et al. (2026), who validated their classifiers on calibration data before applying them to the conversational paradigm.

4.3 Motivational Design Principle

Beauchemin et al. (2024) found that BCI adaptation without motivational context produced no significant learning gains and lower satisfaction than a fixed-pace control — a striking null result with direct design implications. ACCLP therefore embeds motivational scaffolding structurally: each conversational topic has an explicit communicative goal (e.g., 'explain this concept to me as clearly as you can'), providing purpose-driven engagement analogous to the financial incentive used in the AM condition of Beauchemin et al. (2024), but through intrinsic rather than extrinsic means. This is consistent with that study's interpretation that the motivational effect likely transitioned from extrinsic to intrinsic motivation over the course of the task.

5. Technical Architecture

Component	Specification	Empirical Justification
EEG System	64-channel active electrodes, 500 Hz sampling, wireless	Orf et al. (2025): 32-ch mobile EEG yielded reliable TRFs during natural conversation and walking

Real-Time Pipeline	Lab Streaming Layer (LSL) + Python decoder, 250 ms update	Mihic Zidar et al. (2026) and Orf et al. (2025) both used LSL for multimodal synchronization
Cognitive Load Index	Alpha power (8-13 Hz) at parietal electrodes + frontal theta (4-7 Hz); 6-s sliding windows	Beauchemin et al. (2024): P7 alpha index; Mihic Zidar et al. (2026): fbCSP on theta/alpha bands
Classifier	LDA with filter-bank CSP features; within-session calibration	Mihic Zidar et al. (2026): fbCSP-LDA achieved 67-81% accuracy on workload classification
AI Agent	GPT-4o or Claude API with real-time complexity parameter control	Mihic Zidar et al. (2026): OpenAI Realtime API for spoken dialogue; demonstrated feasibility
Speech Interface	Whisper ASR for participant speech; ElevenLabs TTS for AI output	Mihic Zidar et al. (2026): same pipeline used in pilot study
Audio Processing	DNN-based speaker separation; onset envelope extraction	Orf et al. (2025): SpeechBrain DNN separated participant/experimenter speech for EEG analysis

6. Research Questions and Hypotheses

Primary Research Questions

- RQ1: Can frontal theta power and parietal alpha suppression — validated in controlled memory tasks (Beauchemin et al., 2024) and partially in a Spelling Bee spoken task (Mihic Zidar et al., 2026) — reliably distinguish cognitive load levels during sustained naturalistic AI dialogue?
- RQ2: Does EEG-adaptive AI dialogue regulation produce measurably different neural and comprehension outcomes compared to a fixed-complexity sham condition?
- RQ3: Does embedded motivational framing (as suggested by Beauchemin et al., 2024) modulate the effectiveness of closed-loop BCI adaptation in a conversational setting?

Hypotheses

H1 (Neural Validity): Frontal theta will significantly increase and parietal alpha will decrease during high-complexity AI dialogue segments compared to low-complexity segments, replicating the alpha desynchronization pattern confirmed by Beauchemin et al. (2024) and the workload increase trend observed by Mihic Zidar et al. (2026).

H2 (Closed-Loop Efficacy): Participants in the EEG-adaptive condition will maintain a moderate cognitive load zone for significantly longer cumulative duration than in the sham condition, indexed by theta/alpha ratio stability — directly operationalizing the 'Goldilocks Zone' adaptive mechanism from Beauchemin et al. (2024).

H3 (Transfer Validity): The within-session decoder accuracy achieved during Phase 1 calibration will positively predict classifier performance during Phase 2 dialogue, consistent with the moderate cross-condition encoding accuracy correlation ($r = 0.31$) reported by Orf et al. (2025) between focused and natural listening conditions.

H4 (Neural Speech Tracking): Auditory temporal response functions (TRFs) during AI speech reception — the listening phase of dialogue — will show canonical P1-N1-P2 morphology with fronto-central scalp topography, consistent with Orf et al. (2025) who obtained such TRFs during natural conversational listening (encoding accuracy M = 0.025, compared to focused listening M = 0.036).

7. Methodology

7.1 Participants

N = 40 healthy young adults (ages 18-35), right-handed, native English speakers, no history of neurological disorder. Target 50% female. Sample size is justified by the effect sizes reported by Beauchemin et al. (2024) (r = 0.32-0.51 for BCI+Motivation learning gains) and the pilot nature of Mihic Zidar et al. (2026) (N = 2 per condition), which underscores the need for a fully powered study.

7.2 EEG Acquisition and Preprocessing

64 active electrodes, 10-20 montage, sampled at 500 Hz, referenced to linked mastoids. Preprocessing will follow the ICA-based pipeline: bandpass filtering 0.1-45 Hz, automated artifact rejection, and ICA decomposition to remove ocular and muscular artifacts — consistent with the preprocessing applied by Mihic Zidar et al. (2026) using EEGLAB and the AMICA algorithm. Audio from both participant and AI will be recorded and synchronized via the Lab Streaming Layer (LSL), following the multimodal synchronization approach of both Orf et al. (2025) and Mihic Zidar et al. (2026).

7.3 Cognitive Load Index and Adaptive Rules

Following Beauchemin et al. (2024), cognitive load will be estimated using a sliding-window alpha-band power index (6-second windows, updated every 6 seconds), supplemented by frontal theta power as validated in Mihic Zidar et al. (2026). Per-session thresholds will be calibrated from Phase 1 N-back data, identical to the Beauchemin et al. (2024) approach of computing separate 0-back and 2-back averages. The adaptive rules follow a three-class scheme: high load triggers simplification of AI output; low load triggers increased complexity; optimal load yields no change — matching the three-class (0/1/2) logic of the Beauchemin et al. (2024) BCI system.

7.4 Neural Speech Tracking Analysis

Following Orf et al. (2025), temporal response functions (TRFs) will be estimated for AI speech received by the participant during the listening phases of dialogue, using the mTRF toolbox and ridge regression with cross-validation. This provides an independent neural index of auditory processing quality across different AI complexity levels — asking whether higher AI complexity reduces encoding accuracy (as natural conversation reduced encoding accuracy relative to focused listening in Orf et al., 2025).

7.5 Primary Outcome Measures

- **Neural:** Frontal theta (Fz, F3, F4), parietal alpha (Pz, P3, P4), theta/alpha ratio, auditory TRF encoding accuracy during AI speech reception
- **Behavioral:** Comprehension probe accuracy (3 questions per topic block), turn-taking latency, NASA-TLX subjective workload ratings post-block
- **Linguistic:** AI complexity parameter time-series, lexical diversity of participant responses, proportion of time in each cognitive load zone

8. Expected Results and Contributions

If H1 is supported, ACCLP would demonstrate that the alpha desynchronization pattern confirmed in controlled memory tasks (Beauchemin et al., 2024) survives in the richer, more variable context of interactive dialogue — extending the partial evidence from Mihic Zidar et al. (2026) to a properly powered study.

If H2 is supported, ACCLP would provide the first demonstration that the closed-loop BCI Goldilocks zone approach (Beauchemin et al., 2024) can be implemented with a conversational AI agent as the adaptive medium, rather than a screen-based information delivery interface.

If H3 is supported, it would confirm that within-session calibration — as used by both Beauchemin et al. (2024) and Mihic Zidar et al. (2026) — produces decoders that generalize adequately to the target conversational task, establishing a practical pipeline for future neuroadaptive conversational systems.

If H4 is supported, auditory TRF morphology during AI speech reception would serve as an independent marker of comprehension quality across complexity levels, extending the natural-listening TRF findings of Orf et al. (2025) into a fully interactive dialogue context.

9. Limitations and Mitigation Strategies

Limitation	Mitigation Strategy
Agreement/implicit evaluation classifiers do not transfer reliably to continuous dialogue (Mihic Zidar et al., 2026)	ACCLP focuses exclusively on workload — the construct with demonstrated cross-paradigm transfer; agreement is deferred to future work
Single-electrode alpha index misses distributed neural dynamics (Beauchemin et al., 2024 limitation section)	Use 64-channel EEG with fbCSP spatial filters spanning frontal and parietal regions; validate against multi-electrode indices
Encoding accuracy reduced in natural conversation vs. focused listening (Orf et al., 2025: 0.025 vs. 0.036 mean r)	Treat this reduction as an expected baseline; use TRF encoding accuracy as a dependent variable rather than a quality metric
Motivation critically moderates BCI efficacy (Beauchemin et al., 2024: BCI-alone group showed no gains over control)	Embed goal-oriented conversational framing for all participants; include motivation measures as covariates in analyses

Speech production creates muscle artifacts and speech-gait coupling effects in EEG (Orf et al., 2025)	Model participant speech and gait as separate TRF regressors; use DNN-based speaker separation as in Orf et al. (2025)
LLM response latency may disrupt natural conversational flow (Mihic Zidar et al., 2026)	Pre-buffer AI responses; target <1s latency; use streaming token generation

10. References

[1] Beauchemin, N., Charland, P., Karran, A., Boasen, J., Tadson, B., Sénécal, S., & Léger, P.-M. (2024). Enhancing learning experiences: EEG-based passive BCI system adapts learning speed to cognitive load in real-time, with motivation as catalyst. *Frontiers in Human Neuroscience, 18*, 1416683. https://doi.org/10.3389/fnhum.2024.1416683

[2] Mihic Zidar, L., Wicke, P., Bhatia, P., Lutz, R., Klug, M., & Zander, T. O. (2026). Decoding workload and agreement from EEG during spoken dialogue with conversational AI. *arXiv preprint arXiv:2601.05825*. https://arxiv.org/abs/2601.05825

[3] Orf, M., Tune, S., Hannemann, R., & Obleser, J. (2025). Speech, gait, and brain dynamics during natural conversation in motion. *bioRxiv preprint*. https://doi.org/10.64898/2025.12.23.696153

End of Proposal — Adaptive Conversational Cognitive Load Paradigm (ACCLP) | Based exclusively on Beauchemin et al. (2024), Mihic Zidar et al. (2026), and Orf et al. (2025)